

A Three-Compartment Mechanistic Lung Simulator for Mechanical Ventilation Education

Development and Computational Verification of an Inspectable Browser-Based Model

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Version evaluated: v0.4.4.1 | Educational exploration prototype | Not a clinical decision tool

Abstract

Background

Mechanical ventilation is difficult to teach because the relationship among pressure, flow, volume, resistance, compliance, and recruitment is dynamic and often hidden inside commercial simulators. An educational model is most useful when its assumptions are explicit, its internal state is inspectable, and numerical failure is distinguishable from a physiologically impossible request.

Objective

To describe the development and computational verification of a browser-based, three-compartment mechanistic lung simulator designed for education and model exploration, with emphasis on transparent initialization, solver behavior, and failure semantics.

Methods

The simulator represents the lung as three quasi-static tissue compartments connected through a common central airway resistance and compartment-specific branch resistances. Each compartment follows a finite-capacity nonlinear elastic pressure-volume relationship and may undergo discrete recruitment or derecruitment. Patient-like mechanical parameters are separated from scenario-dependent choices such as initial PEEP, recruitment state, and ventilator settings. The numerical solver uses an analytic Jacobian, scaled convergence criteria, and direction-aware boundary checks. Computational verification included initialization, Jacobian accuracy, resistance-compliance behavior, conservation laws, recruitment constraints, volume- and pressure-controlled ventilation, timestep convergence, failure semantics, and step-level solver instrumentation.

Results

The v0.4.4.1 release passed 127 prespecified acceptance assertions across 21 test files with no failures. Multi-breath simulations were stable across all four mechanical phenotypes in both volume-controlled and pressure-controlled modes. Timestep refinement from 2 ms to 1 ms to 0.5 ms showed monotone convergence in the tested scenarios. Failed steps preserved time and state, and the simulator distinguished boundary infeasibility from solver non-convergence. In an illustrative high-recruitability scenario at PEEP 5 cmH₂O and a 1 ms timestep, 10 seconds of simulated time completed in approximately 1.5 seconds with no solver failures.

Conclusions

This release establishes computational consistency and transparent failure behavior within the tested domain. It does not establish physiological accuracy, clinical validity, or educational

effectiveness. The current system is best described as a mechanistic lung simulator rather than a patient-specific digital twin.

Keywords: mechanical ventilation; simulation; lung mechanics; recruitment; numerical methods; medical education; computational verification

Introduction

Mechanical ventilation requires learners to reason simultaneously about airway pressure, flow, lung volume, resistance, compliance, time constants, and recruitment. These relationships are familiar in isolation but become harder to interpret when several mechanisms interact over successive breaths. Simulation can make those interactions visible, but only if the model is sufficiently transparent for the learner - and the reviewer - to understand what the software is assuming and why it behaves as it does.

Educational lung simulators span physical test lungs, high-fidelity mannequins, and software models. Web-based systems offer an additional advantage: they can expose state variables, equations, and solver diagnostics directly to the user and can be distributed without dedicated hardware [1-3]. That transparency also creates a methodological obligation. A model that produces plausible waveforms but silently guesses missing state, violates conservation, or obscures numerical failure can teach the wrong lesson while appearing credible.

The present work therefore focuses on computational verification rather than clinical validation. We developed a browser-based simulator that represents the lung as three mechanically distinct compartments and explicitly separates intrinsic model parameters from scenario-dependent clinical choices. We then tested whether the implementation behaves consistently with its own equations, preserves state when a numerical step fails, and differentiates an impossible requested boundary condition from ordinary solver non-convergence.

The principal contribution of this release is not a new clinical model of acute respiratory distress syndrome (ARDS). The four available phenotypes are deliberately labeled as mechanical constructs - Reference, Low-recruitability, Moderate-recruitability, and High-recruitability - and are not mapped to Berlin ARDS severity categories or to clinical cohorts. The goal is a reproducible educational substrate on which later physiologic calibration and educational validation can be built.

Methods

Model architecture

The simulator uses three quasi-static lung compartments representing mechanically distinct regions of lung tissue. Each compartment has its own nominal capacity, elastic stiffness, resistance, perfusion fraction, dead-space fraction, and recruitment state. The compartments share a central airway resistance before flow divides into the branch pathways. This topology allows the model to express heterogeneous filling and emptying while retaining a compact state space that can be inspected in real time.

For each open compartment, volume is related to pressure by a finite-capacity exponential elastic law. As compartment volume approaches its available capacity, the slope of the pressure-volume

relationship steepens. The analytic derivative of this relationship is used directly by the Newton solver rather than estimated numerically. Closed compartments are constrained to zero volume and zero recruitment.

Separation of phenotype and scenario

A central design decision was to separate parameters intended to describe the mechanical phenotype from parameters that define a particular simulation scenario. The phenotype contains tissue and airway properties; the scenario contains the starting state and ventilator choices. In particular, initial PEEP and initial recruitment state are supplied explicitly by the caller. If either is missing, initialization fails rather than substituting a hidden default. This change was introduced after earlier development versions showed how silently inferred recruitment could propagate through downstream comparisons.

Component	Stored in phenotype	Supplied by scenario
Mechanical properties	Compartment fractions, resistance, capacity, elastic stiffness, perfusion/dead-space fractions	-
Shared airway	Central airway resistance; airway opening pressure	-
Initial state	-	Initial PEEP; initial recruitment state
Ventilator	-	Mode, FiO2, PEEP, respiratory rate, tidal volume or pressure target, flow/timing settings

Mechanical phenotypes

Four prespecified mechanical phenotypes are included. Their names refer to the size of the recruitable pool and are intended only to provide reproducible test conditions. They are not clinical diagnostic categories.

Internal identifier	Display label
phenotype_baseline	Reference phenotype
phenotype_low_recruitability	Low-recruitability phenotype
phenotype_moderate_recruitability	Moderate-recruitability phenotype
phenotype_high_recruitability	High-recruitability phenotype

Numerical solution and convergence

At each mechanics step, the solver determines the change in compartment volumes together with the common branch pressure while enforcing the requested total flow. Newton iteration is driven by an analytic Jacobian. Residuals are normalized to bounded volume and pressure reference

scales so that convergence tolerance does not become artificially permissive in large-capacity states. The implementation also uses line search and adaptive subdivision when a full step cannot be accepted.

If Newton iteration cannot find an acceptable solution, the simulator performs a direction-aware feasibility check. During inspiration, the requested incremental volume is compared with remaining available capacity. During expiration, requested removal is compared with the gas volume that can actually be removed. A step that exceeds these necessary bounds is classified as `INFEASIBLE_BOUNDARY`; otherwise the failure is classified as `SOLVER_NONCONVERGENCE`. The nonlinear solve remains authoritative - the boundary checks are used to interpret failure, not to replace the solver.

Failure semantics and instrumentation

A failed mechanics step is transactional: simulation time does not advance and the partially computed state is not committed. Per-step diagnostics record Newton iterations, subdivision depth, line-search halvings, active-set transitions, residual measures, and convergence status. These diagnostics were included so that a user or reviewer can distinguish a difficult but converged step from one that failed or required substantial numerical intervention.

Computational verification

Verification was organized as prespecified acceptance tests rather than retrospective examples. The battery addressed the model implementation at several levels: local constitutive behavior, network conservation, multi-breath controller behavior, timestep sensitivity, and explicit failure contracts. Tests were executed in CommonJS mode under Node v26.8.2 using the package test runner.

Domain	What was tested
Initialization	Pressure-consistent initialization; closed-compartment behavior; lower volume bound; rejection of missing recruitment state
Constitutive law	Analytic Jacobian compared with a numerical derivative
Single-compartment dynamics	Small-signal time constant approximately equal to resistance x tangent compliance
Network physics	Central and branch flow conservation; pressure drop across central resistance
Resistance limit	Peak-to-plateau pressure difference approaches zero as resistance decreases
Recruitment constraints	Closed-compartment invariants and derecruitment feasibility projection
Multi-breath simulation	Volume-control and pressure-control stability across all four mechanical phenotypes
Numerical convergence	Monotone behavior with timestep refinement at 2, 1,

	and 0.5 ms
Failure behavior	State preservation on failed steps; separation of infeasible boundary conditions from non-convergence
Instrumentation	Presence and validity of per-step solver statistics, including active-set transitions

Results

Acceptance testing

All 127 acceptance assertions passed across 21 test files; no assertion failed. The passing set covered initialization and state contracts, analytic derivatives, small-signal dynamics, conservation, resistance limits, recruitment behavior, multi-breath operation, timestep convergence, failure semantics, and solver instrumentation.

Verification section	Assertions passed
Initialization and state contract	11/11
Analytic Jacobian	3/3
Small-signal dynamics	1/1
Conservation and airway topology	8/8
Low-resistance limit	2/2
Recruitment invariants	4/4
Multi-breath VC/PC stability	5/5
Timestep convergence	3/3
Failure semantics	7/7
Solver instrumentation	4/4

The remaining package-level tests covered additional controller, gas-exchange, metric, and determinism behavior; together they produced the total of 127 passing assertions.

Multi-breath behavior and timestep refinement

Volume-controlled and pressure-controlled simulations completed across all four mechanical phenotypes without solver failure in the designated acceptance scenarios. Timestep refinement from 2 ms to 1 ms to 0.5 ms produced monotone convergence in the tested volume-control and pressure-control cases. These results support numerical consistency of the current implementation within the tested operating domain; they do not establish that the underlying parameter choices reproduce human physiology.

Failure classification

The failure contract behaved as intended. When a step failed, the simulator preserved both simulation time and total lung volume rather than committing a partial update. Tests separately exercised impossible inspiratory filling and impossible expiratory emptying, confirming that both could be classified as boundary infeasibility. Feasible expiration did not trigger the infeasibility classifier. This distinction is important for an educational simulator because “the model cannot converge” and “the requested state cannot exist within the model” are conceptually different events.

Illustrative performance benchmark

Performance was measured as an engineering benchmark rather than a correctness criterion. In the High-recruitability phenotype with PEEP 5 cmH₂O and a 1 ms timestep, 10 seconds of simulated time required approximately 1.5 seconds of wall-clock time. The run contained 10,000 mechanics steps, 5,859 Newton iterations, and no solver failures. Approximately 31.9% of steps required bounded subdivision, and nine active-set transitions were recorded.

Metric	Observed value
Simulated duration	10 s
Mechanics timestep	1 ms
Mechanics steps	10,000
Wall-clock time	~1.5 s
Total Newton iterations	5,859
Mean Newton iterations per step	0.59
Subdivision-bounded steps	31.92%
Active-set transitions	9
Solver failures	0

A separate three-breath instrumentation test observed three active-set transitions across 6,924 mechanics steps. That test exists to verify that the transition counter is connected and exercised; the observed transition frequency is not presented as a physiological quantity.

Discussion

This work describes a deliberately modest milestone: a mechanistic lung simulator that can explain its own numerical behavior. The principal result is not that the model is clinically realistic, but that its internal rules are consistently applied and testable. That distinction matters in simulation research. Before asking whether a model represents patients, it should first be possible to show that the software represents the model it claims to implement.

Three design choices appear especially important for educational use. First, separating phenotype from scenario prevents ventilator choices or hidden starting conditions from masquerading as

patient characteristics. Second, the analytic Jacobian and scaled convergence rules make solver behavior more reproducible across operating regimes. Third, explicit failure semantics preserve state and provide a reason for failure rather than allowing an unstable or impossible step to contaminate subsequent breaths.

The three-compartment structure is intentionally interpretable. It can represent regional heterogeneity in resistance, elastic behavior, gas capacity, perfusion, dead space, and recruitment without requiring a high-dimensional anatomical model. This makes it useful for demonstrating mechanisms such as unequal time constants, pressure-flow relationships, and recruitment-dependent capacity. At the same time, the simplicity of the architecture means that many clinically important phenomena are absent or highly abstracted. A passing software test cannot compensate for missing physiology.

The term “digital twin” is therefore not used for the current release. In clinical and engineering literature, a digital twin generally implies a model linked to a specific physical system through patient- or device-specific data and ongoing calibration. The present simulator is neither patient-specific nor calibrated to measured bedside data. Reserving that terminology avoids implying a level of clinical personalization that has not been demonstrated.

Limitations

Several limitations are fundamental. The mechanical phenotypes were chosen as inspectable educational constructs rather than fitted to human ARDS cohorts. The selected stiffness values and recruitability patterns are didactic parameters, not validated population estimates. Gas exchange and other downstream metrics included in the software are outside the scope of this computational-verification report. No expert face-validity study, learner study, or comparison with clinical waveforms is presented here.

The active-set transition mechanism is instrumented but only sparsely exercised in the benchmark scenario. A more explicit active-set or boundary formulation is planned for later development. Historical-pressure initialization is also not implemented; callers must currently provide an explicit initial recruitment state. Finally, browser rendering and user-interface behavior were not part of the acceptance battery reported here.

Implications and next steps

The next stage should move from software verification toward validation in two separate directions. Physiologic validation would require prespecified comparisons between simulator outputs and measured or well-characterized experimental or clinical data. Educational validation would require usability and learning-outcome studies in the intended learner population. Those studies should be treated as distinct from the numerical verification presented here.

Conclusion

Version 0.4.4.1 provides an executable, inspectable three-compartment lung simulator with reproducible evidence of computational consistency in its tested domain. The release explicitly separates mechanical phenotype from simulation scenario, uses an analytic and scaled nonlinear solution strategy, preserves state on failed steps, and distinguishes boundary infeasibility from solver non-convergence. These features provide a transparent foundation for education and future

model development. They do not establish clinical realism, patient-specific validity, or educational effectiveness.

Software and Reproducibility

The simulator, source code, test suite, and versioned release artifacts are available from the project website and public source repository. The evaluated release is v0.4.4.1. The test suite contains 21 files and 127 assertions and is executed with the package test runner. Machine-readable test results, numerical diagnostics, and the performance benchmark are generated with the release artifacts.

Project website: <https://vent.defying-logic.com/ards-twin-v0.4/>

Source repository: <https://github.com/DrFillster/medical-vent-twin/tree/main/ards-twin-v0.4>

Versioned release: <https://github.com/DrFillster/medical-vent-twin/releases/tag/v0.4.4.1>

Disclosures

Ethics approval: Not applicable to the computational development work described here; no patient-specific data were used.

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Conflicts of interest: [Add details here].

Author contributions: [Add details here].

Data availability: The computational artifacts described above are publicly available with the versioned release.

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